

Decomposition Using Multivalued Attribute (4NF) -- DBMS

Definition: A relation is in Fourth Normal Form (4NF) if it is in BCNF and for every non-trivial Multivalued Dependency (MVD) $X \twoheadrightarrow Y$, X is a Super Key.

1. Detailed Technical Explanation

Multivalued Dependency (MVD):

A multivalued dependency $X \twoheadrightarrow Y$ (read as "X multidetermines Y") exists in schema R if the presence of a pair of tuples (t_1, t_2) with $t_1[X] = t_2[X]$ implies that there must also exist tuples t_3 and t_4 combining X, Y, and the remaining attributes $Z = R - (X \cup Y)$.

MVD Symbol: $X \twoheadrightarrow Y$ (X independent multidetermines Y)

Example: 4NF Decomposition

Relation STUDENT_INFO(Student_ID, Mobile_No, Skill)

Data Tuple Redundancy (UNF/BCNF Violation):

101	9876543210	Java
101	9876543210	Python
101	9123456789	Java
101	9123456789	Python

(Note: Adding 1 new skill for student 101 requires inserting 2 new rows because Mobile_No and Skill are independent!)

4NF Decomposition Algorithm:

Decomposed 4NF Tables:

2. Core Concepts & Memory Keywords

* Multivalued Dependency (MVD): Independence between two multi-valued attributes associated with the same key.

3. Must-Write Points for Exams

* 4NF handles independent multi-valued attributes that BCNF cannot resolve.

4. Quick Recall Flow

BCNF Table -> Identify Independent Multi-Valued Attributes ($X \twoheadrightarrow Y$) -> Split into $R_1(X, Y)$ & $R_2(X, Z)$ -> 4NF Achieved